

Tutorial 2

Sets and Systems of linear equations

1. Let $\mathcal{U}$, $\mathcal{L}$ and $\mathcal{D}$ respectively be the set of all upper triangular, all lower triangular and all diagonal 3×3 matrices, and let I be the 3×3 identity matrix. For each of the following statements say whether it is true or false, and give a reason for your answer.

- (a) $I \in \mathcal{D}$
- (b) $\{I\} \in \mathcal{D}$
- (c) $\{I\} \subseteq \mathcal{U}$
- (d) $\emptyset \subseteq \mathcal{D}$
- (e) $\mathcal{D} = \mathcal{U} \cap \mathcal{L}$
- (f) $\{A^T : A \in \mathcal{U}\} \subseteq \mathcal{U}$
- (g) $\{A^T : A \in \mathcal{D}\} = \mathcal{D}$
- (h) $\{A^T : A \in \mathcal{U}\} = \mathcal{L}$

2. Decide whether the following statement is true or false:

“For any two sets A and B , at least one of the following statements is true:

- (a) $A \subseteq B$ (b) $B \subseteq A$ (c) $A = B$.”

If the statement is true, prove it. If it is false, give a counter-example.

3. Determine which of the following statements are true:

- (a) The empty set is a subset of every set.
- (b) If $A \subseteq B$, then $A = B$.
- (c) If $A = B$, then $A \subseteq B$.
- (d) Since $\emptyset$ is a member of $\{\emptyset\}$, $\emptyset = \{\emptyset\}$.
- (e) The set of all symmetric 3×3 matrices and the set of all skew-symmetric 3×3 matrices are disjoint.

4. Let $A = (a_{ij})$ be a 3×3 matrix **in row echelon form**. Consider the equation

$$A \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

- (a) Is it possible for this equation to have:

(i) A unique solution (ii) No solution (iii) Infinitely many solutions?
 Give an example of A in each possible case. If you think some case(s) cannot occur, explain why not.

(b) Find a condition, in terms of only the diagonal entries (the a_{ii}) of A , that distinguishes between the cases (i) and (iii) above.

5. Four augmented matrices are given below with their associated variables. Are any of these augmented matrices **not** in row-echelon form? If so, which one(s)?

For the ones that are in row-echelon form:

- (i) say which variables are free and which are dependent;
- (ii) solve the system associated with the augmented matrix and write down the **solution set**. Use the **vector form** of the solution to write down the solution set (i.e. write the solution set as a set of vectors).

$$(a) \left(\begin{array}{ccc|c} 0 & 1 & 1 & 4 \\ 0 & 0 & 1 & 1 \end{array} \right) \quad x, y, z$$

$$(b) \left(\begin{array}{ccc|c} 2 & 1 & 1 & 4 \\ 0 & 3 & 1 & 1 \\ 0 & 1 & 0 & 0 \end{array} \right) \quad x, y, z$$

$$(c) \left(\begin{array}{ccccc|c} 1 & 1 & 1 & 1 & 1 & 8 \\ 0 & 0 & 1 & 1 & 1 & 8 \\ 0 & 0 & 0 & 5 & 0 & 0 \end{array} \right) \quad x, y, z, w, v$$

$$(d) \left(\begin{array}{ccccc|c} 4 & -1 & 3 & 7 & 1 & 2 \\ 0 & 0 & 3 & 4 & 9 & 1 \\ 0 & 0 & 0 & 0 & 0 & 5 \end{array} \right) \quad x_1, x_2, x_3, x_4, x_5$$

6. Sketch the following set:

$$\{(x, n) \in \mathbb{R} \times \mathbb{N} : 1 \leq x < 5 \text{ and } 0 \leq n \leq x\}.$$